Which of the following characterize or are included in gas-liquid chromatography?

I. Gas mobile phase

II. Liquid stationary phase

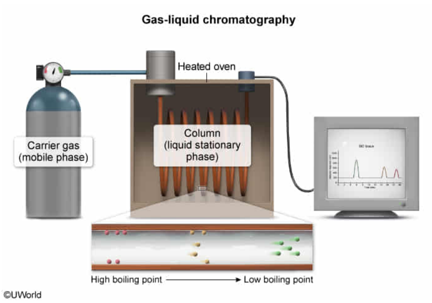
III. Separation based only on polarity

IV. Room-temperature column conditions

- A. I only [3%]
- ✔ B. I and II only [53%]
- C. I and III only [11%]
- D. I, II, and IV only [31%]

Correct
52% answered correctly

Explanation:



Gas-liquid chromatography is a technique used to separate molecules in a mixture based on their boiling points. A gas chromatograph consists of an injection port, mobile and stationary phases, a column in a heated oven, a detector, and a computer for data analysis. The **mobile phase** is an **inert gas** such as helium or nitrogen **(Number I)**, and the **stationary phase** is a **liquid** that coats a solid support on the inside of the column **(Number II)**.

A small amount of a liquid mixture is injected into the gas chromatograph, and the compounds in the mixture are vaporized by heating. The vapors then travel through the **column** in a **heated oven** to the detector. The most volatile molecules (ie, low boiling points) spend more time in the gas phase than they spend interacting with the stationary phase of the column, so they rapidly migrate to the detector. However, molecules with higher boiling points condense more readily and spend more time interacting with the liquid stationary phase. These molecules make their way through the column slowly as the temperature in the oven increases.

(Number III) In gas-liquid chromatography, mixture components are separated primarily based on *boiling point*. Although the boiling point is determined by several factors (eg, polarity, intermolecular forces, molecular weight), each of these factors is already accounted for by the boiling point itself. For molecules with the same number of carbons, the boiling point increases with the increasing strength of intermolecular forces and relative polarity. However, a sufficiently large nonpolar molecule can have a higher boiling point than a small polar molecule. For example, a nonpolar 20-carbon alkane chain has a higher boiling point than a polar 4-carbon carboxylic acid (343 °C vs 164 °C) and would move through the column more slowly despite being relatively nonpolar.

(Number IV) The column must be placed in a heated environment to allow volatile molecules to remain in the gas phase. Columns kept at room temperature would not separate the molecules.

Educational objective:
Gas-liquid chromatography is a technique that separates compounds in a mixture based on boiling point. The mobile phase is an inert gas, and the stationary phase is a liquid that coats a column in a heated oven. Molecules with lower boiling points reach the detector before molecules with higher boiling points.

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